

Research Proposal: eRisk 2026 - Task 1

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- Main Proposal Idea: Lead DS@GT ARC project on the eRisk 2026's Task 1, conversational depression detection, and submit a working notes paper at the CLEF 2026 conference.
- Challenge Reference: [CLEF eRisk 2026 Task 1](#)

Objective

Design and evaluate a system, that, given limited-turn conversations with LLM personas can:

- Automatically estimate the persona's depression severity following BDI-II standards (total score + severity level per category)
- Detect which BDI-II symptoms are present
- Operate under time constraints and handle each newly released persona model with short interaction window

The final goal is a set of reproducible runs to the eRisk 2026 lab and a working notes paper documenting methodology, results, and lessons learned.

Background and Motivation

eRisk is a long-running evaluation campaign, part of the broader CLEF lab network, for early risk detection in mental health using text and interaction data. The 2026 Task 1 focuses on LLM personas that simulate different levels and profiles of depression according to the BDI-II questionnaire, and systems are evaluated on symptom detection and overall severity estimation.

The key motivation is to move from static text classification toward interactive, clinically informed interviewing strategies. Instead of passively scoring existing posts, our system must actively query personas, extract structured symptom evidence, and make severity predictions. Such methods are relevant to future digital mental health tools where automated agents may assist clinicians by summarizing risk signals from brief conversations.

Key Strategy Summary

1. Treat task as semi-structured “clinical interview” with fixed number of turns (there is prompt limit)
2. Scripted but adaptive question policy that targets all 21 BDI-II symptoms
3. Encode conversations with transformer model (e.g. RoBERTa) with simple temporal context (e.g. question + persona reply + past 1-2 exchanges)

Conversational Policy

- Write question templates that target each depressive symptom (e.g. "How has your sleep been lately?" for the sleeping pattern BDI-II symptom)
 - Avoid asking the persona directly about depression—personas are designed to deflect such questions. Instead, infer symptoms through natural conversational cues (language, tone, expressed thoughts).
 - Consult faculty in the School of Psychology on prompt wording, conversational focuses, and relative BDI-II symptom importances to ground our question design in clinical expertise.
- Start with broad questions to get general answers, and narrow down based on responses (adaptive decision-tree structure)
 - If positive for a targeted BDI-II symptom, continue probing that domain with deeper follow-up questions; otherwise, pivot to other symptoms.
 - Cap turns per BDI-II symptom to a hard limit; make prediction once the threshold is hit.
 - Explore **algorithmic question policies** and **adaptive LLM interviewer prompts** to compact more information per question and efficiently cover all 21 symptoms within the turn budget.

Modeling

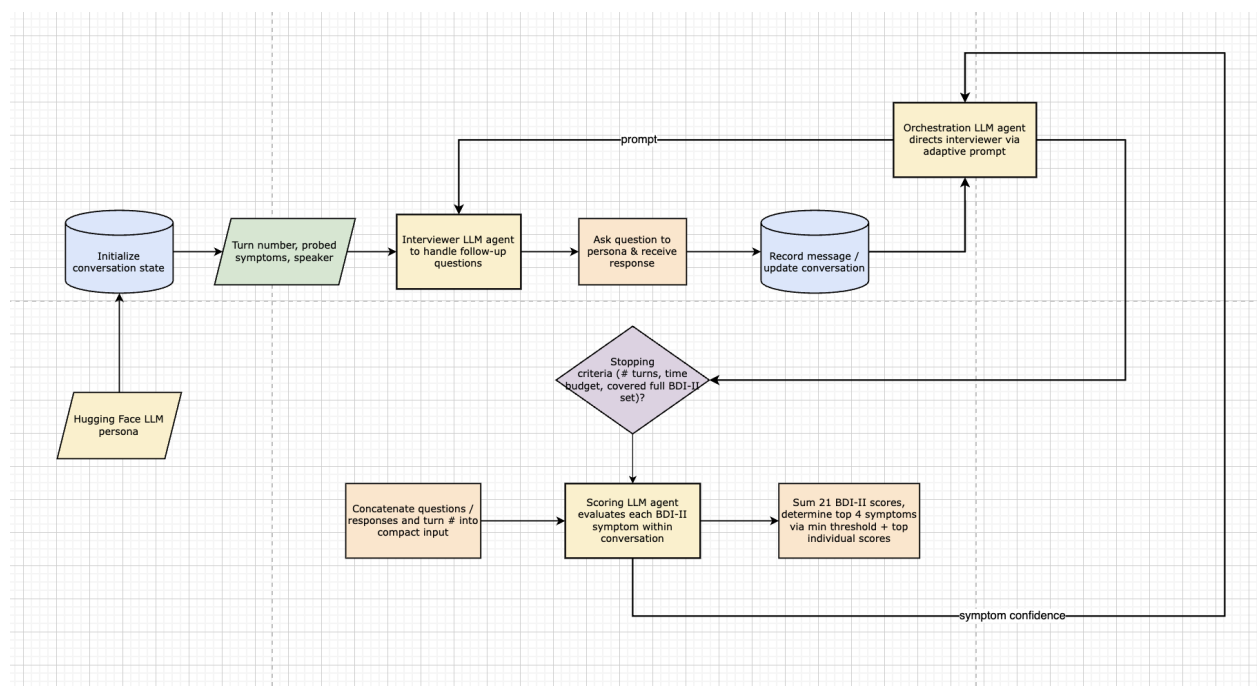
- **Multi-agent pipeline:** Separate the interview and scoring responsibilities across distinct LLM agents.
 - **Conversation agent** conducts the interview with the persona
 - Prompt-engineered to ask natural, open-ended questions that elicit symptom-relevant information without directly asking about depression
 - Compound questions that touch multiple symptoms simultaneously (e.g. "How's a typical day been going — energy, motivation, that kind of thing?") to maximize information per turn.
 - **Scoring agent** independently analyzes the conversation transcript after each turn
 - Structured, per-symptom rubric prompts (e.g. "0 = no indication of sleep disruption, 1 = mild complaints, 2 = significant disruption, 3 = severe/pervasive") with chain-of-thought reasoning to score each of the 21 BDI-II symptoms individually, rather than producing a single holistic score.
 - Tracks confidence level for each symptom to guide orchestration & conversation agent on which areas to explore deeper
 - **Orchestration agent** sits between the two
 - Reads the scorer's per-symptom confidence estimates after each turn
 - Identifies which symptoms are underexplored or ambiguous
 - Directs the conversation agent on where to probe next or when to move on
 - Closes the feedback loop so the interview adapts in real time rather than following a static question list.
- **Ensemble / multi-pass scoring:** Run the scoring agent multiple times per turn (varied prompts or temperature > 0) and aggregate via median to reduce variance and improve reliability.
- **Confidence-based early stopping:** After each turn, the orchestration agent checks whether all 21 symptoms have crossed a confidence threshold. Once sufficient evidence is gathered, the conversation ends and predictions are submitted, optimizing for the eRisk early-decision penalty.

Exploratory Data Analysis (EDA)

- **BDI-II scoring vs. conversation length (turns):** Examine how predicted scores shift as conversations progress—identify whether additional turns yield diminishing returns or improved accuracy.
- **Consistency assessment:** Interview each persona 5 times independently and plot the resulting BDI-II scores to assess score stability and persona consistency.
- **Human baseline comparison:** Engage in manual conversations with personas, produce educated-guess BDI-II scores, and compare against automated AI predictions to gauge system performance and identify failure modes.

MVP: A quick MVP can simply be hardcoded questions, and then training a transformer model to output 21 symptom scores through full interview content instead of message by message.

Process Diagram (WIP)



End-to-end Task

Our system conducts structured interviews with LLM personas to detect depression severity via BDI-II scoring. It uses a semi-structured question policy combined with an interviewer LLM agent to probe 21 BDI-II symptoms and a separate scoring agent to evaluate each symptom given the full

conversation content. Per-symptom scores are summed for BDI-II, categorized into severity bands, and the top 4 symptoms are recorded in the output.

We submit fully automated runs to eRisk 2026 Task 1, demonstrating robust conversational symptom detection via a CLEF 2026 Working Notes paper.

Timeline

First Model Release: February 16, 2026

Last Model Release: April 20, 2026 (releases 2 per week, 20 total)

Run Submission: May 7, 2026

Notebook Paper Submission: May 28, 2026

Notebook Paper Notification: June 30, 2026

Notebook Camera Ready: July 6, 2026

CLEF 2026: September 21, 2026

Infrastructure

Code hosted via DS@GT ARC (add github link). Cloud storage and compute provisioned through Google Cloud Platform with team/project billing.

Collaboration and Supervision

Work coordinated within Georgia Tech's Data Science ARC team. DS@GT regularly participates and publishes top-ranked CLEF working notes. The supervising faculty member will handle registration and assessment respective to course requirements.

References

eRisk 2026. CLEF Lab Website. <https://erisk.irlab.org>

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